

TrES-5b

R-1680

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190610 · created 2026-08-02T11:31:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458644.9013 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.055
Transit depth (Rp/Rs)^2	3.22 +/- 1.96 [%]
Orbital Inclination (inc)	83.55 +/- 2.85
Airmass coefficient 1 (a1)	17.49 +/- 16.51
Airmass coefficient 2 (a2)	-2.16 +/- 1.29
Scatter in the residuals of the lightcurve fit is	141.11 %
Best Comparison Star	#3 - [155, 28]
Optimal Aperture	1.43
Optimal Annulus	14.33
Transit Duration (day)	0.067 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.18 $\pm$ 0.055 vs 0.142 (deviation 0.43 σ)
Duration fit vs published	0.067 d vs 0.0758 d (deviation 0.24 σ)
Mid-time O-C	-6.7 min
Depth SNR	2.0
Residual scatter	141.11 %

Light curve

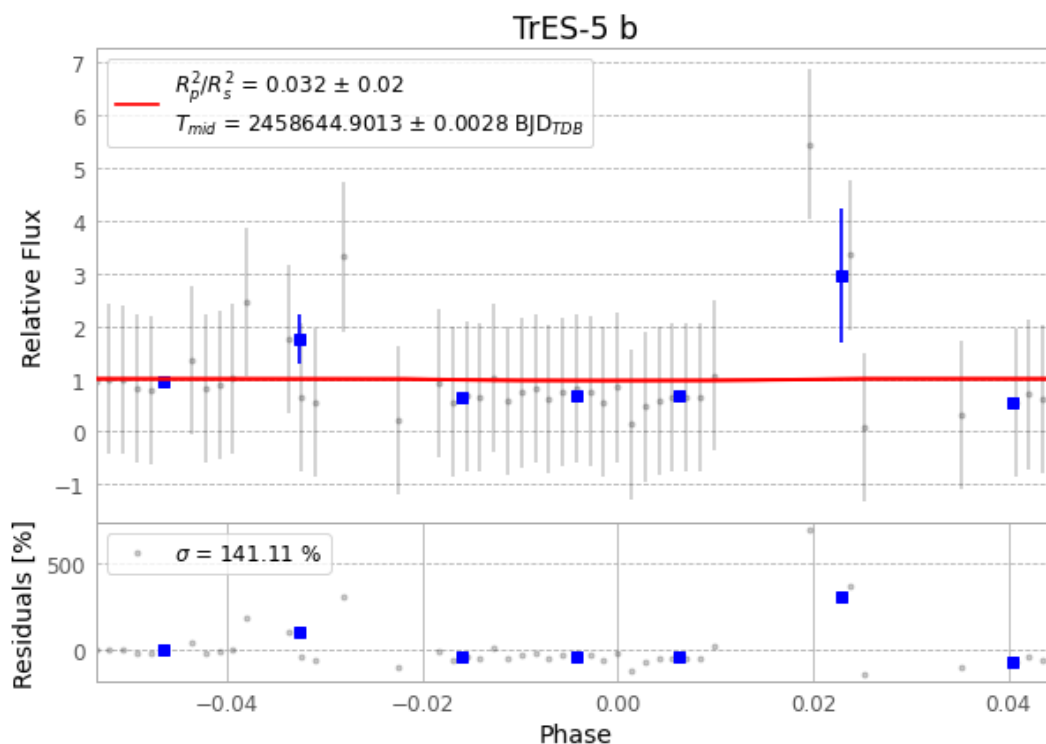


Figure 1 — FinalLightCurve_TrES-5 b_2019-06-10.png (SHA-256 ad3c33f7b04b52cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [451, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e2111710d4c3d8c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), TRES-5190610074212.FITS → TRES-5190610111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190610.log (4118ca6d9d17...), FinalLightCurve_TrES-5 b_2019-06-10.png (ad3c33f7b04b...), FinalLightCurve_TrES-5 b_2019-06-10.csv (b3feb7266c5b...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:31Z.